Repurposing Approved Drugs for Zika Virus

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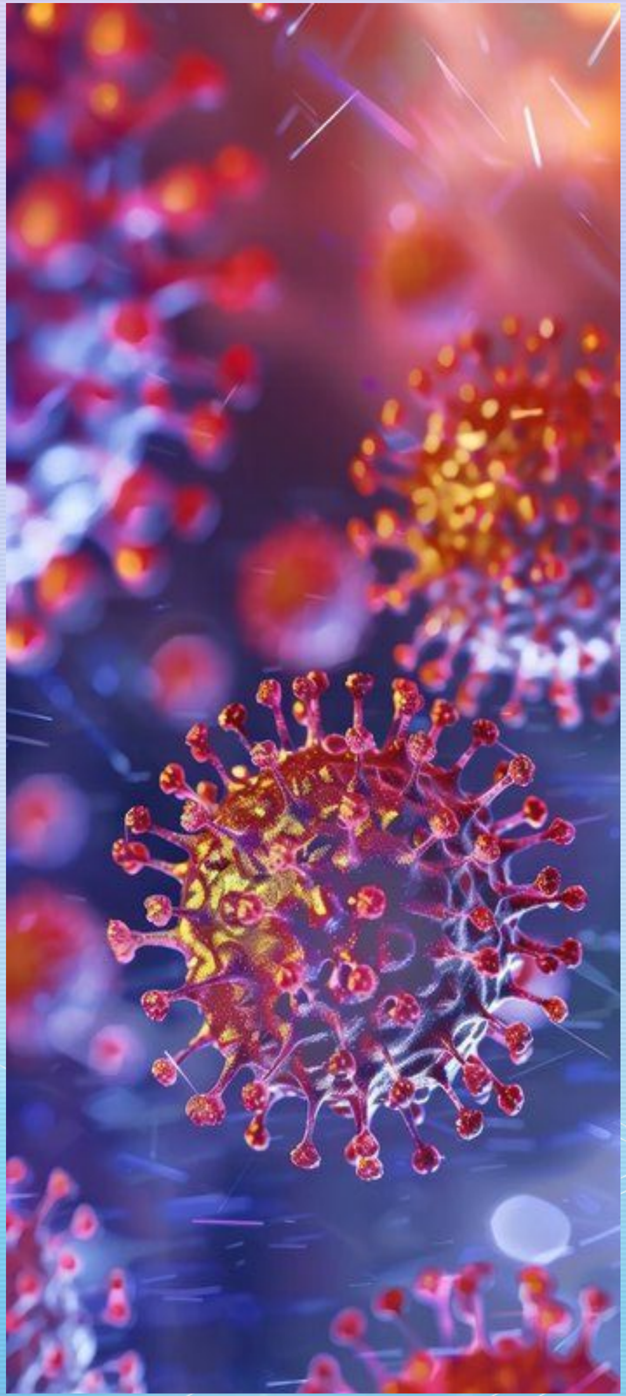
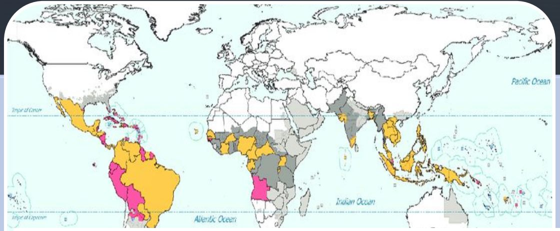


ZIKA VIRUS OVERVIEW

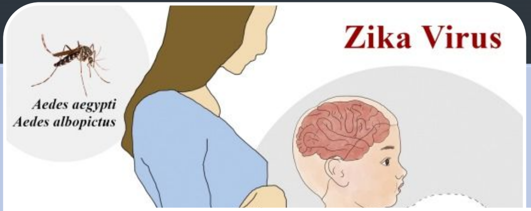
ORIGIN

- Zika virus is a mosquito-borne flavivirus, first identified in monkeys in Uganda in 1947 and later in humans in 1952.
- It is a Positive single-stranded RNA virus



Global Impact

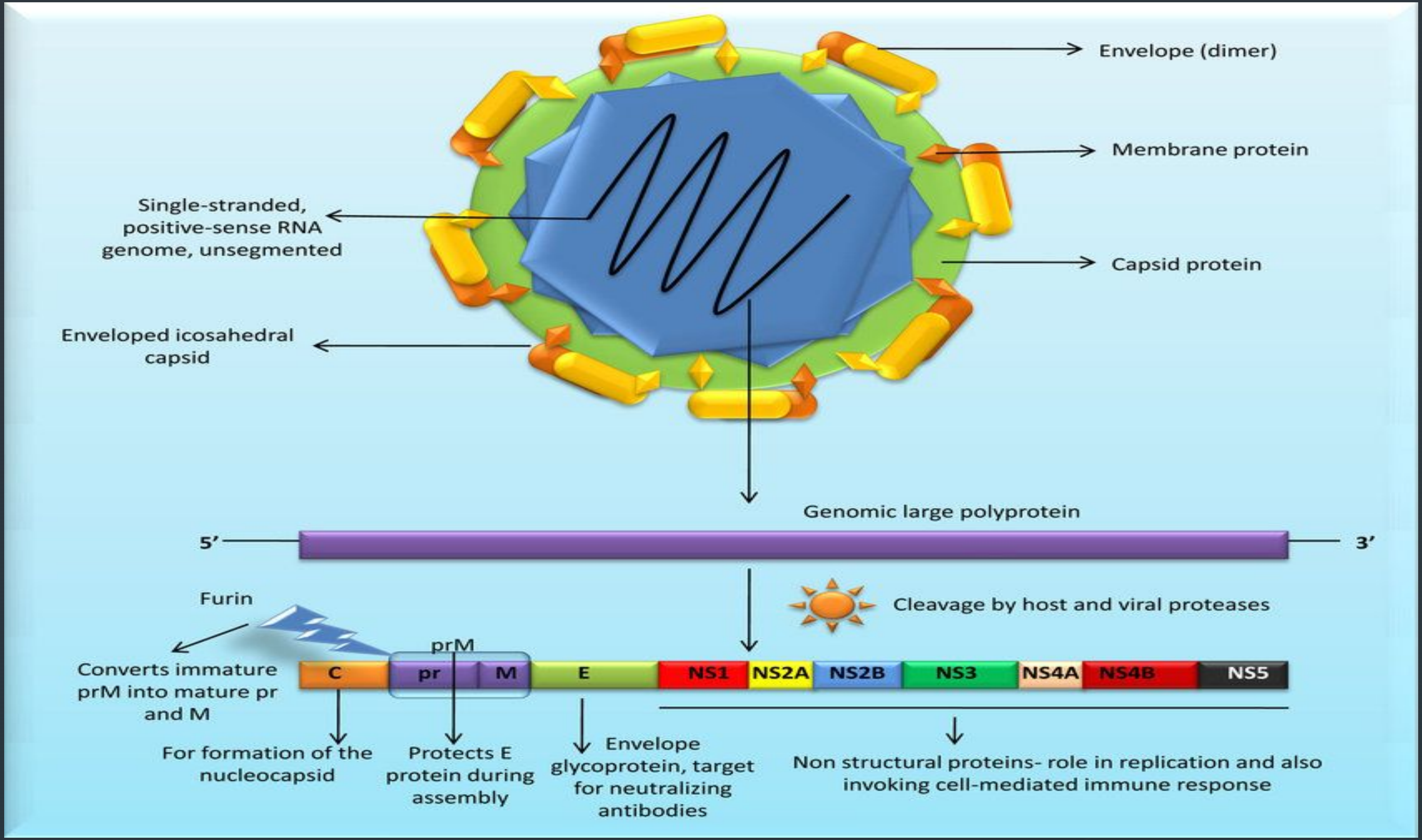
- Zika virus outbreaks have occurred in various regions, including the America, Asia, and the Pacific region of the world, leading to public health concerns.
- WHO declared Zika-related microcephaly a Public Health Emergency of International Concern (PHEIC) in Feb, 2016
- No specific treatment available
- No vaccine available for prevention



Transmission and Symptoms

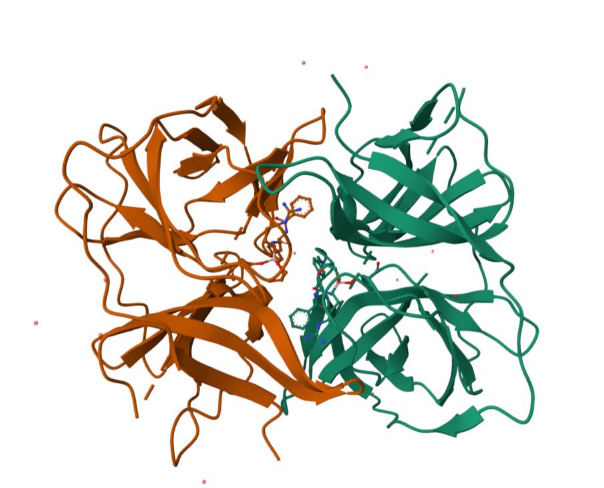
- The virus is primarily transmitted through the bite of infected Aedes mosquitoes.
- Infection symptoms: rash, fever, conjunctivitis, muscle and joint pain
- Infection during pregnancy can cause infants to be born with microcephaly and other congenital malformations as well as preterm birth and miscarriage

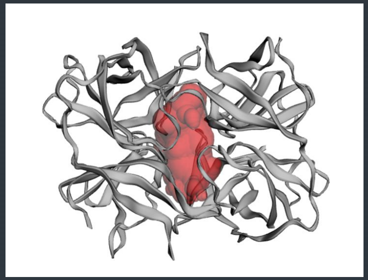
GENOME OF ZIKA VIRUS



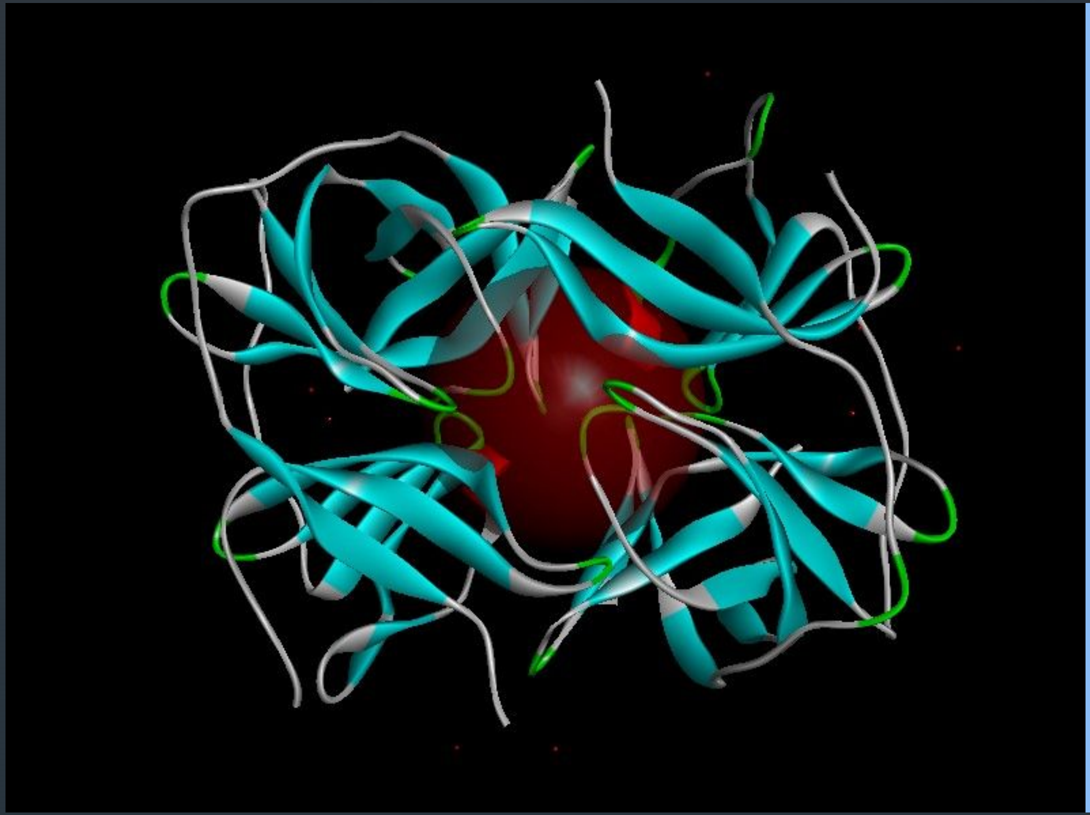
Drug Target- NS2B-NS3 serine protease

- NS3 serine protease is a crucial enzyme that is essential for the replication of the virus.
- NS3 serine protease in association with NS2B performs its autocleavage and cleaves the polyprotein at dibasic sites
- The multiple roles played by the NS2B–NS3 protein in the virus life cycle makes it an attractive target for antiviral drug discovery. (Luo D et.al., 2015 Jun)
- No human homolog
- Non-structural protein
- Druggable
- Availability of crystal structure(PDB ID: 5LC0)





BINDING SITE



- Largest Surface Area
- Largest Volume
- Includes the residues mentioned in UniProtKB/SwissProt as binding site and active site residues.
- Druggability Probability Predicted by POCKDRUG - 0.69

TOP I: Dofequidar

PubChem CID: 213040
Libdock Score: 177.642

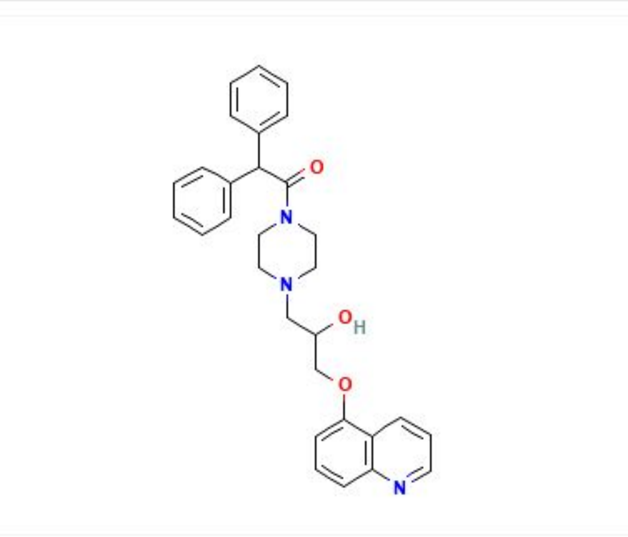
- Dofequidar is an orally-available synthetic quinoline derivative with multidrug resistance modulating properties.
- Dofequidar shows promise as a P-glycoprotein inhibitor for cancer treatment, there is limited information available on its effectiveness against viral infections, including Zika virus.

Dofequidar's mechanism of action (inhibiting P-gp) might not be directly beneficial against viruses itself. P-gp is efflux pump protein, and its main function is to remove unwanted substances from cells. Viruses, however, generally hijack the host cell's machinery for replication. So, inhibiting P-gp might not directly hinder the virus's ability to replicate.

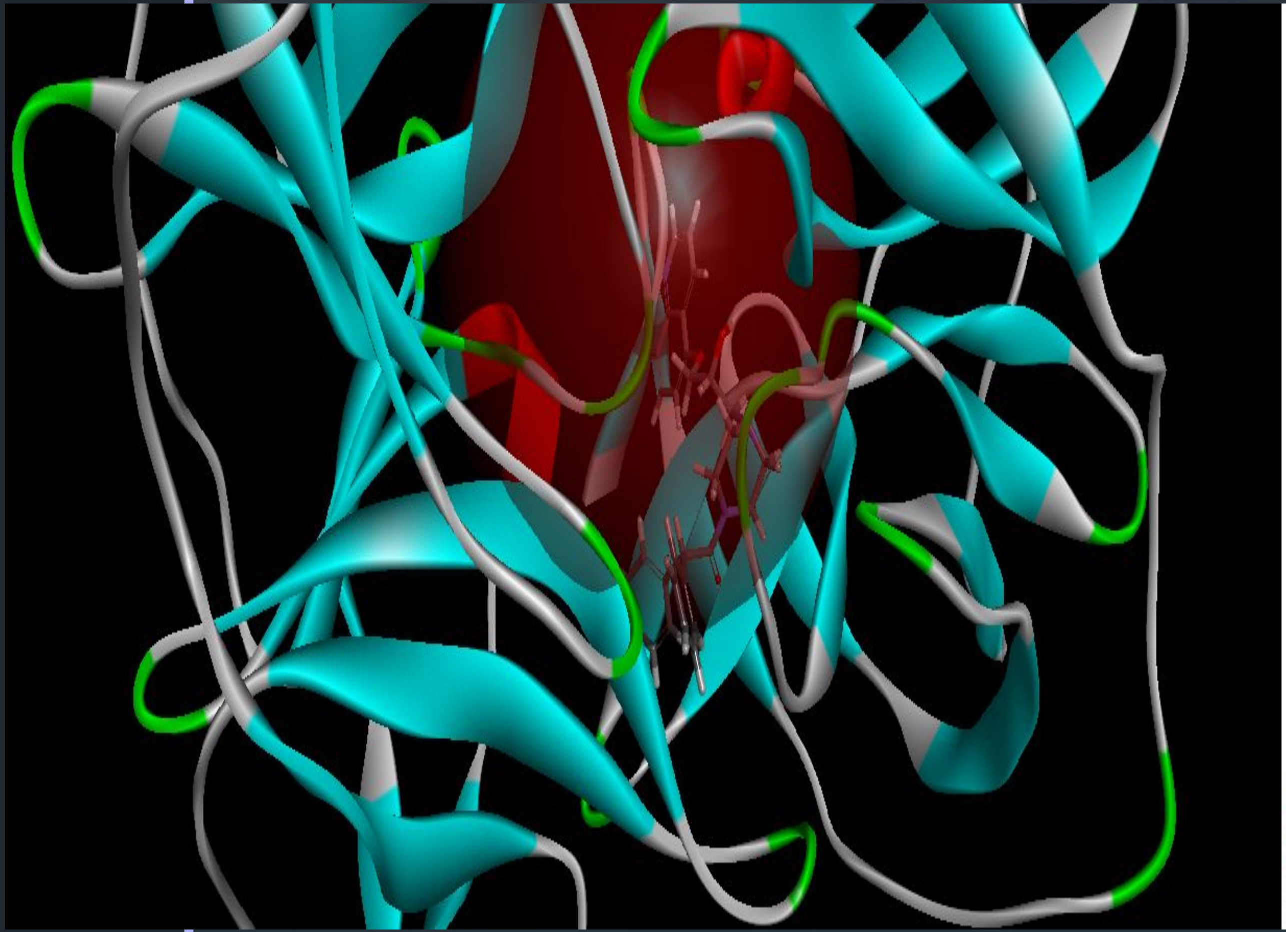
Indirect effect on viruses: Inhibiting P-gp might:

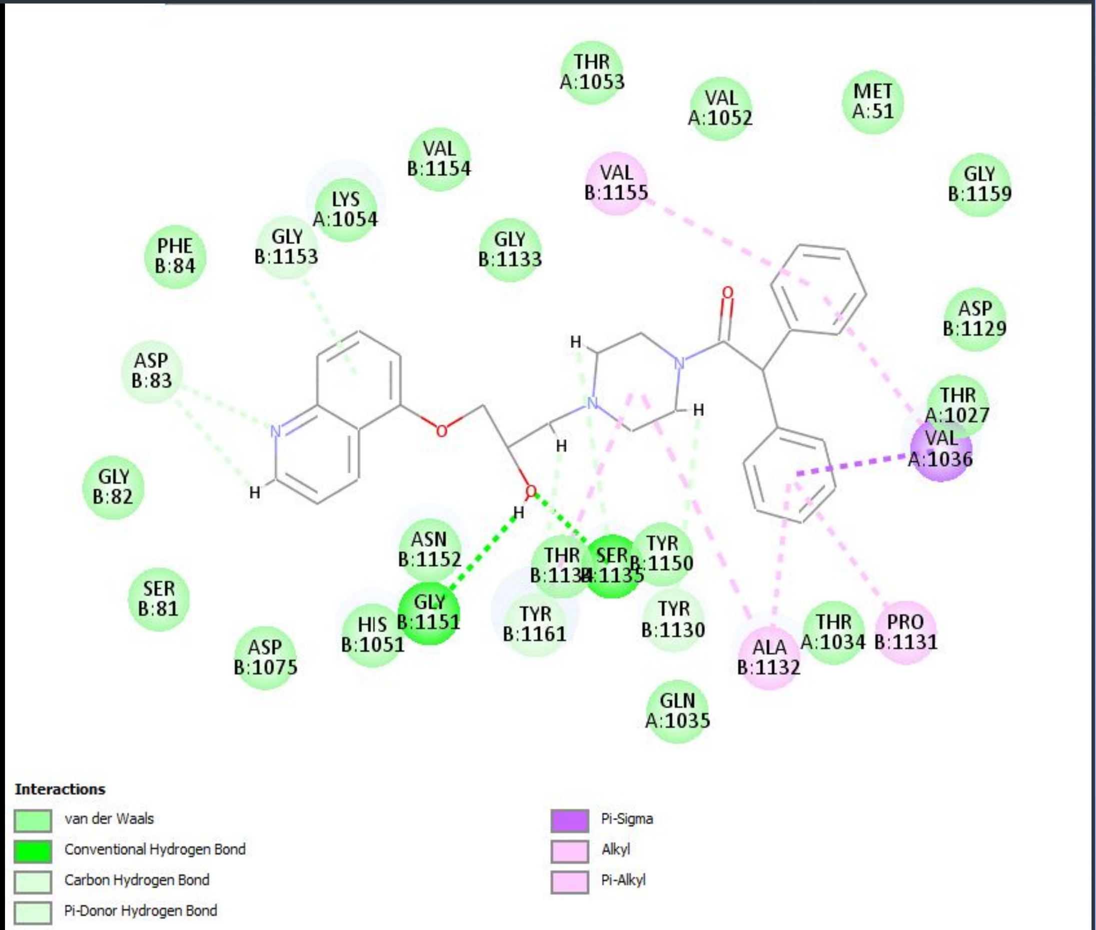
Increase intracellular drug concentration: If an antiviral drug gets pumped out by P-gp, inhibiting P-gp could allow the drug to stay inside the cell for a longer duration, potentially increasing its effectiveness.

Trap viral components: Some viruses might rely on P-gp for exporting certain components they don't need. Inhibiting P-gp could potentially trap those components inside the virus, hindering its functionality.



Interaction of compound and serine protease N3 residues:





II:

1-[2-(3-Biphenyl)-4-methylvaleryl)]amino-3-(2-pyridylsulfonyl)amino-2-propanone

PubChem CID: 5288593

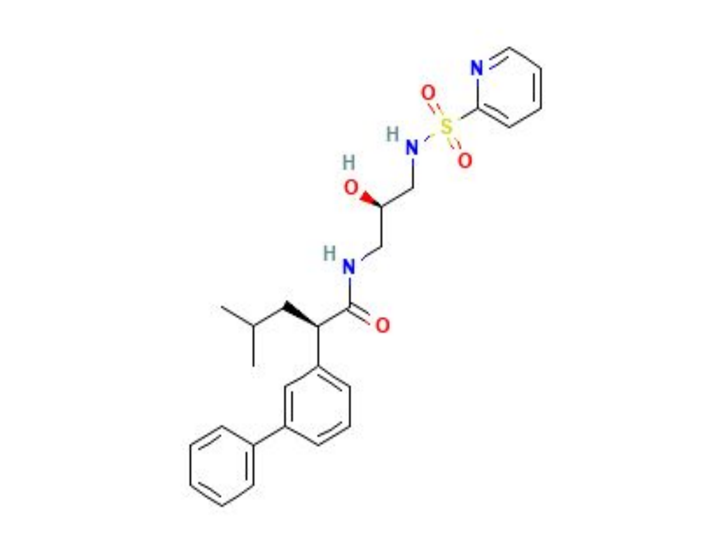
LibDock Score: 176.212

A COVALENT PEPTIDOMIMETIC INHIBITOR

This compound belongs to the class of organic compounds known as biphenyls and derivatives. These are organic compounds containing to benzene rings linked together by a C-C bond.

<u>Mechanism of action:</u> CATHEPSIN K

Cathepsin K is a protease enzyme. Some studies suggest Zika might utilize cathepsins, including Cathepsin K, to enter host cells. Inhibiting Cathepsin K activity could potentially hinder this entry process, reducing viral spread.



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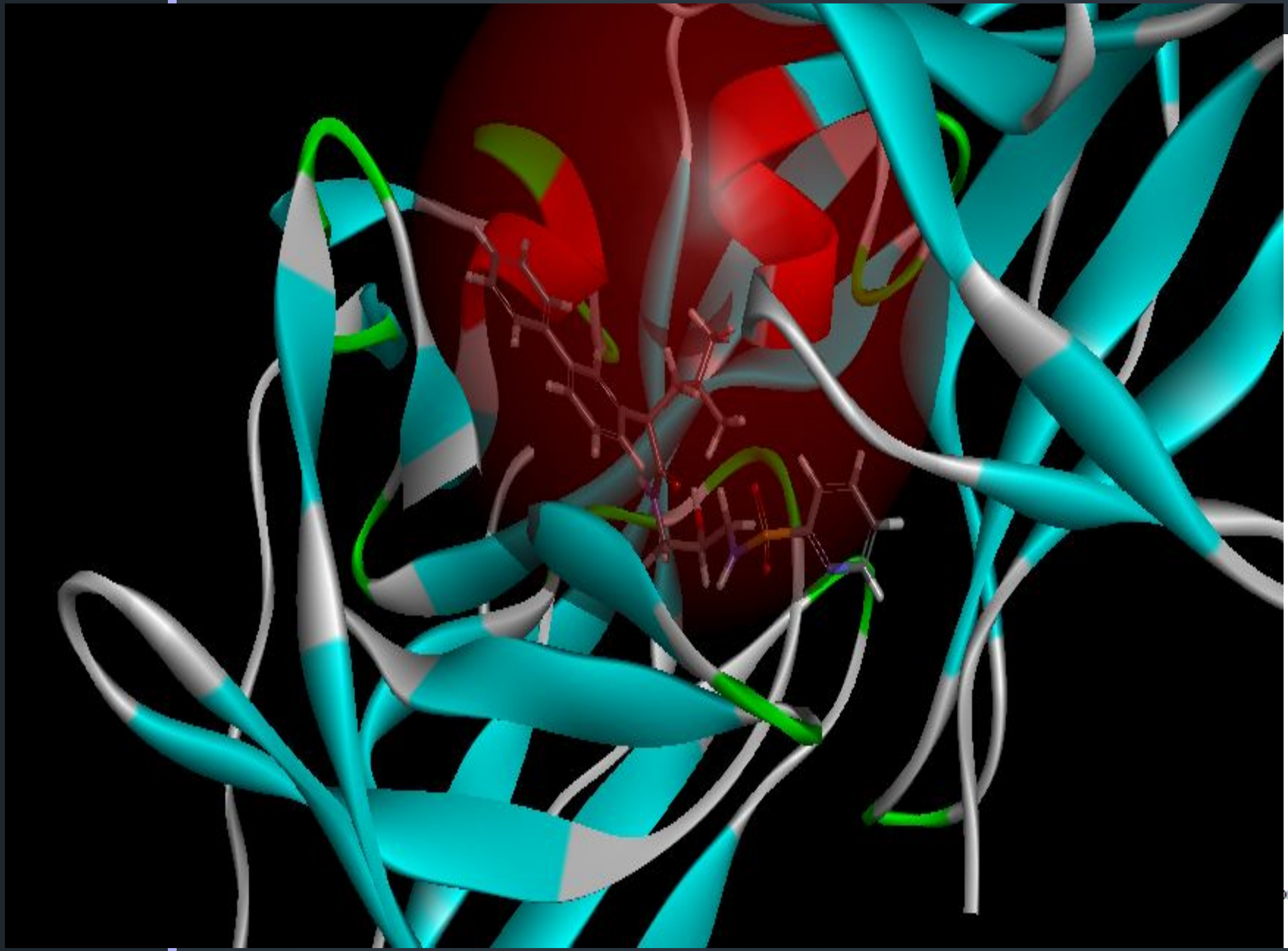
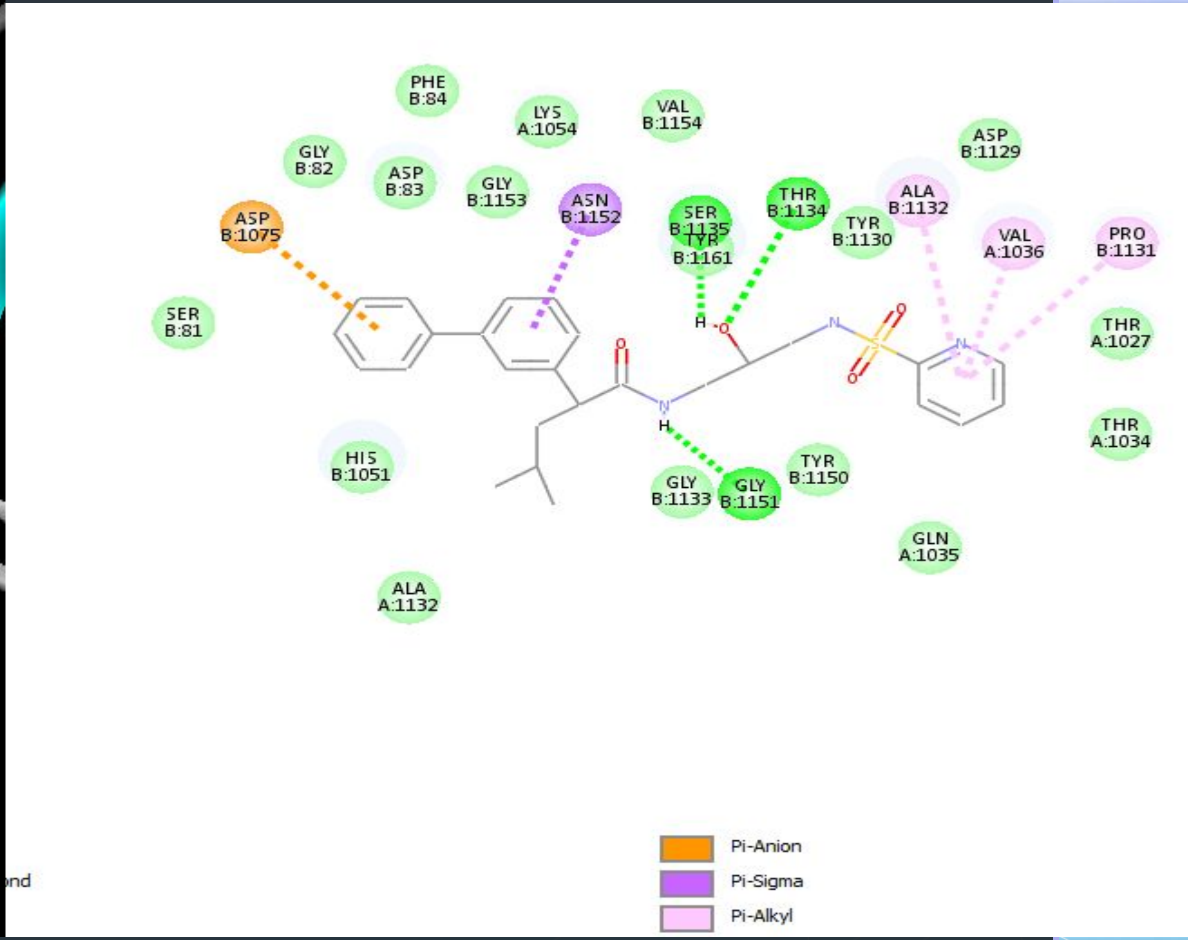




III 10-Propargyl-5,8-dideazafolic acid

PubChem CID: 135438608

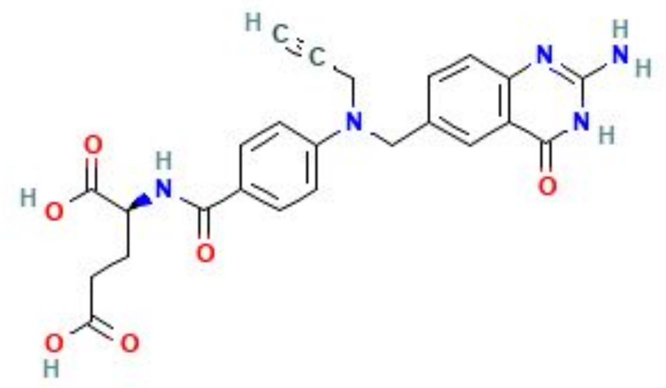
LibDock Score: 173.777

General Function : Thymidylate synthase activity

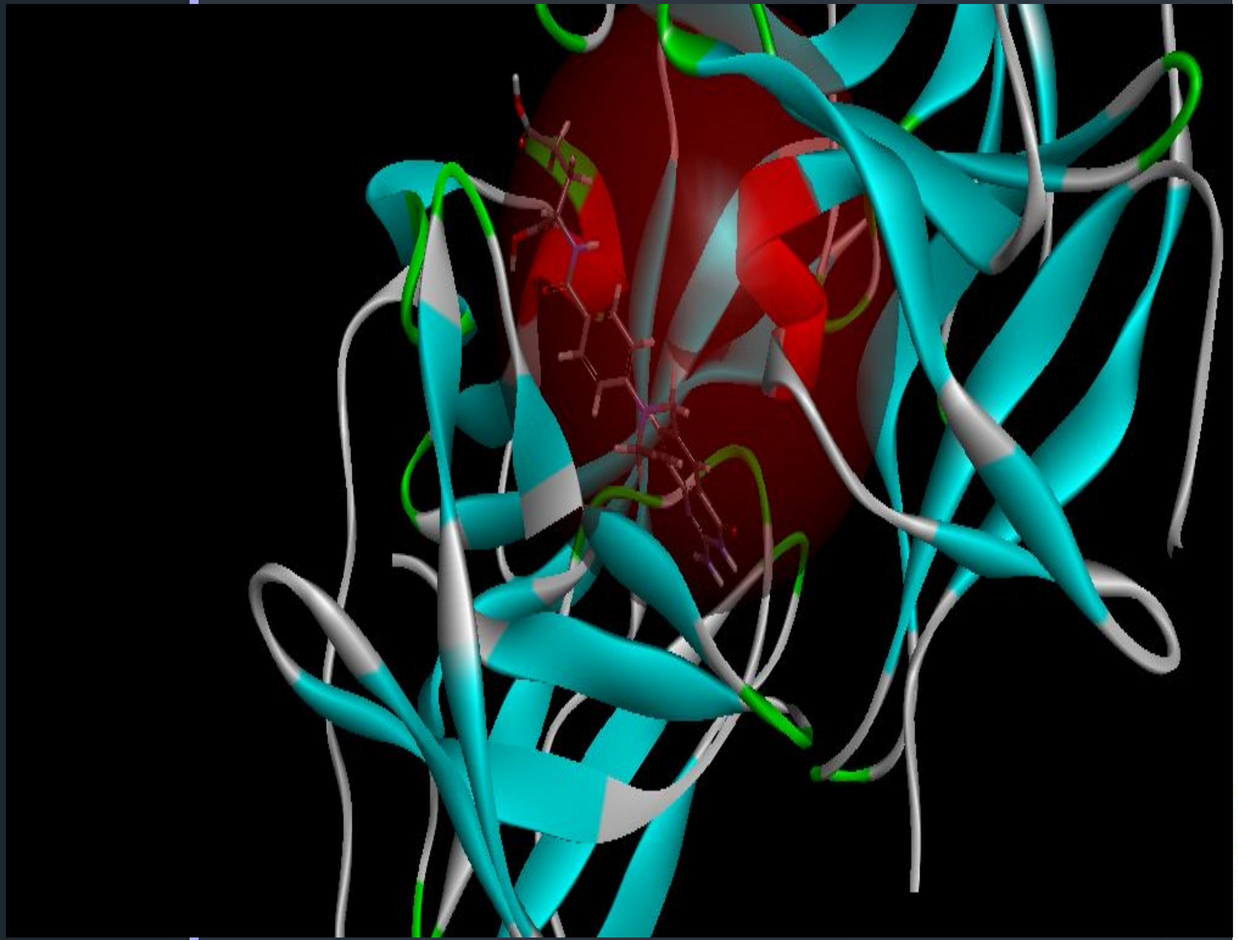
Specific Function: Contributes to the de novo mitochondrial thymidylate biosynthesis pathway.

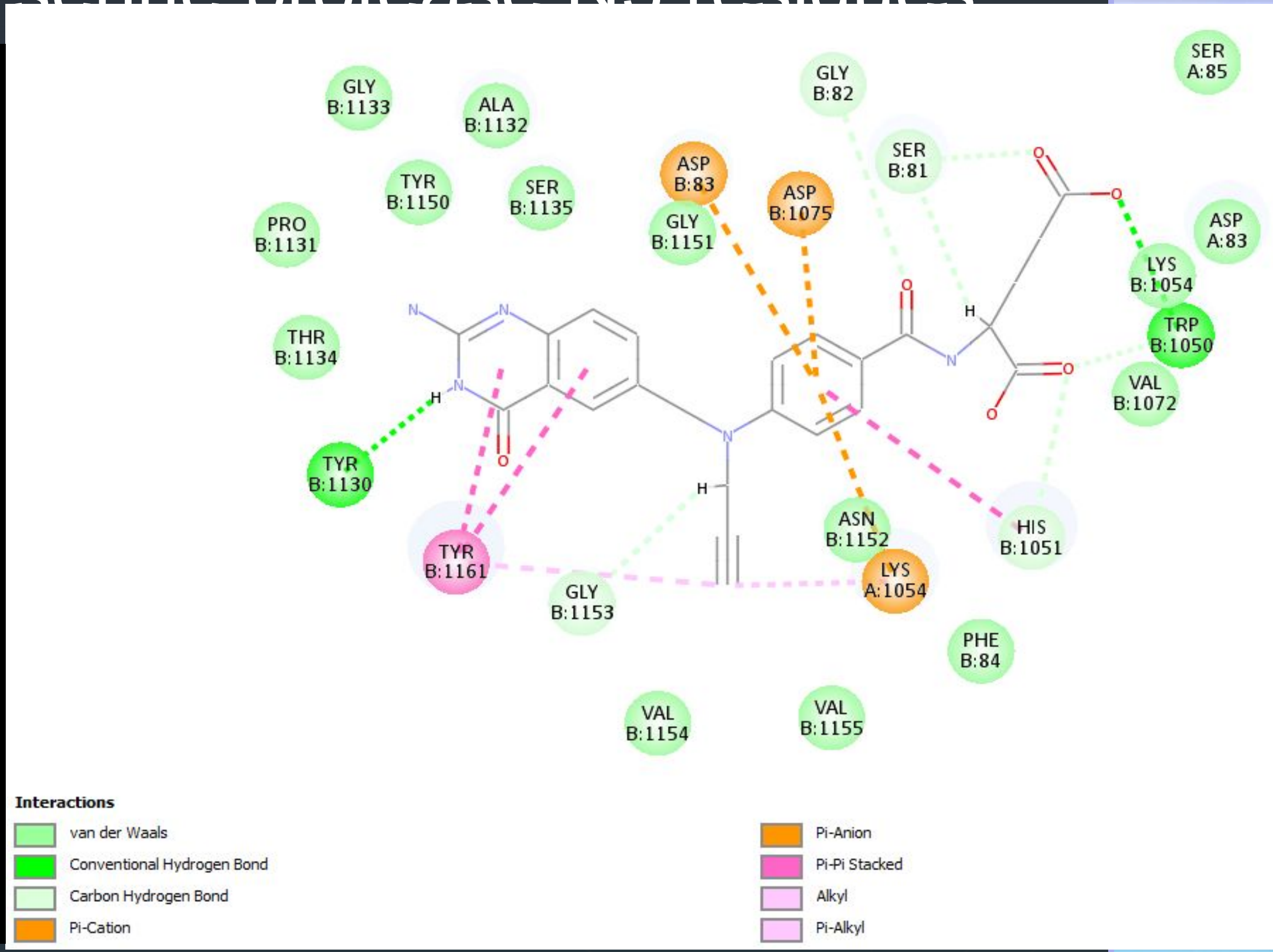
There is no direct relationship but there might be an indirect connection to consider:

While TS wouldn't directly target the Zika virus itself, a healthy and functional TS enzyme might be necessary for the host cell to repair the DNA damage caused by the Zika infection. By providing the necessary building blocks for repair, a functional TS system could potentially help the cell recover from the damage and limit the negative effects of the infection.



Interaction of compound and serine protease N3 residues:

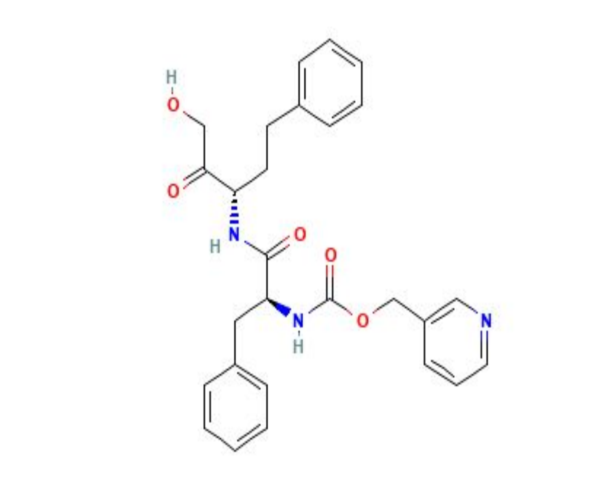




IV -[1-(3-Hydroxy-2-oxo-1-phenethyl-propylcarbamoyl)2-phenyl-ethyl]-carbamic acid pyridin-4-ylmethyl ester

PubChem CID: 5289091

LibDock Score: 173.718



Mechanism of action- **Target** -UCruzipain

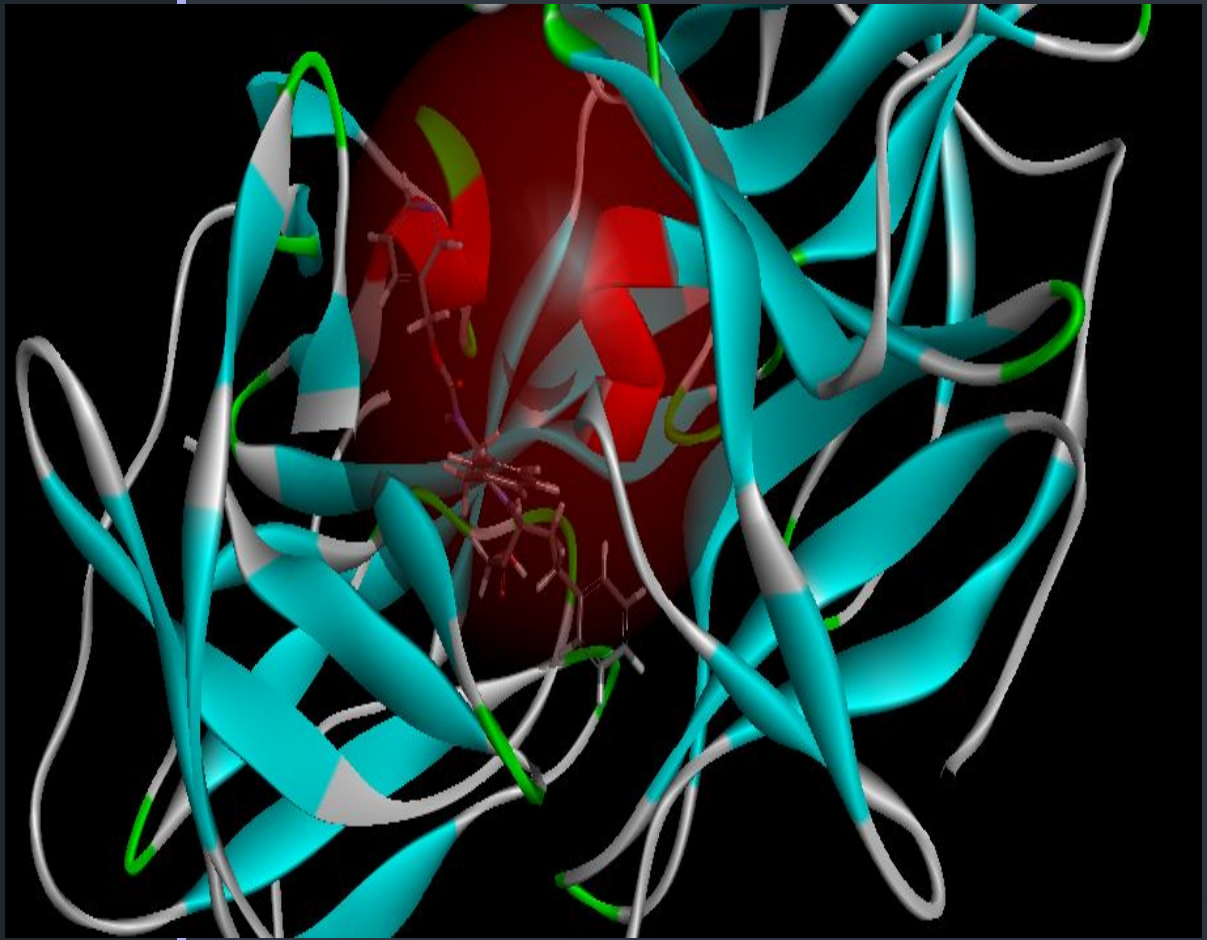
General Function :Cysteine-type endopeptidase activity

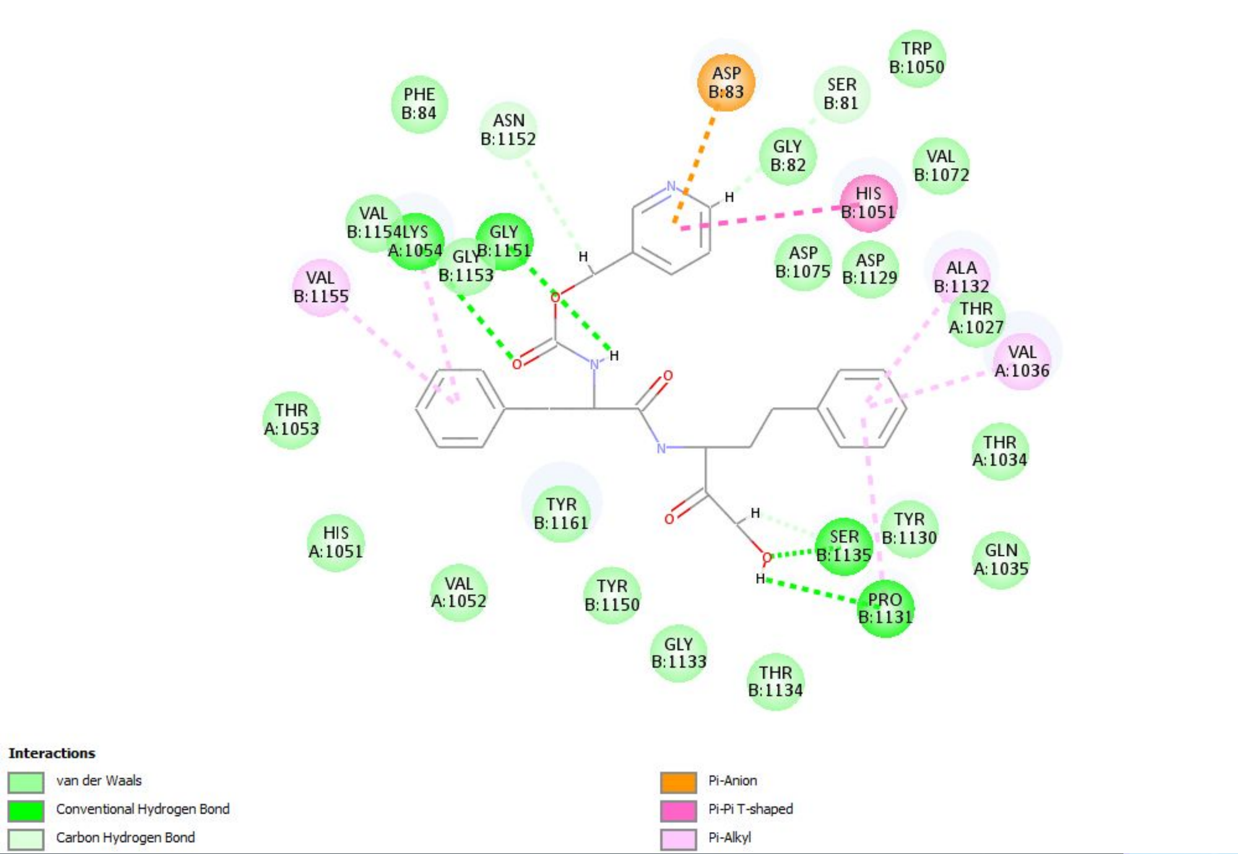
Specific Function :Hydrolyzes chromogenic peptides at the carboxyl Arg or Lys; requires at least one more amino acid, preferably Arg, Phe, Val or Leu, between the terminal Arg or Lys and the amino-blocking group.

Ucruzipain has a similar structure to host cell proteases, it wouldn't directly target the Zika virus itself. Inhibiting Ucruzipain might indirectly affect viral entry by affecting similar host proteases, but more research is needed to confirm this connection.

Specificity challenges: Developing inhibitors specific to host cell proteases involved in Zika entry is crucial. Broadly targeting all cysteine proteases with Ucruzipain inhibitors could have unintended consequences.

Interaction of compound and serine protease N3 residues:





TOP V: Ximelagatran

PubChem CID: 9574101

LibDock Score: 171.468

General Function :Thrombospondin receptor activity

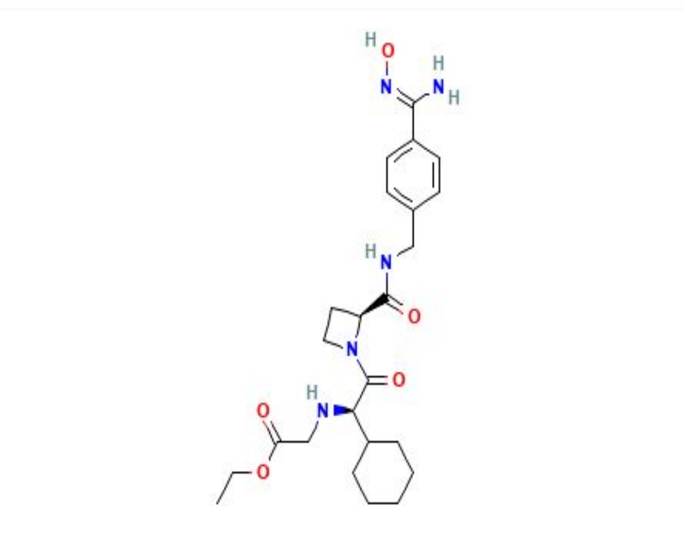
Specific Function

Thrombin, which cleaves bonds after Arg and Lys, converts fibrinogen to fibrin and activates factors V, VII, VIII, XIII, and, in complex with thrombomodulin, protein C.

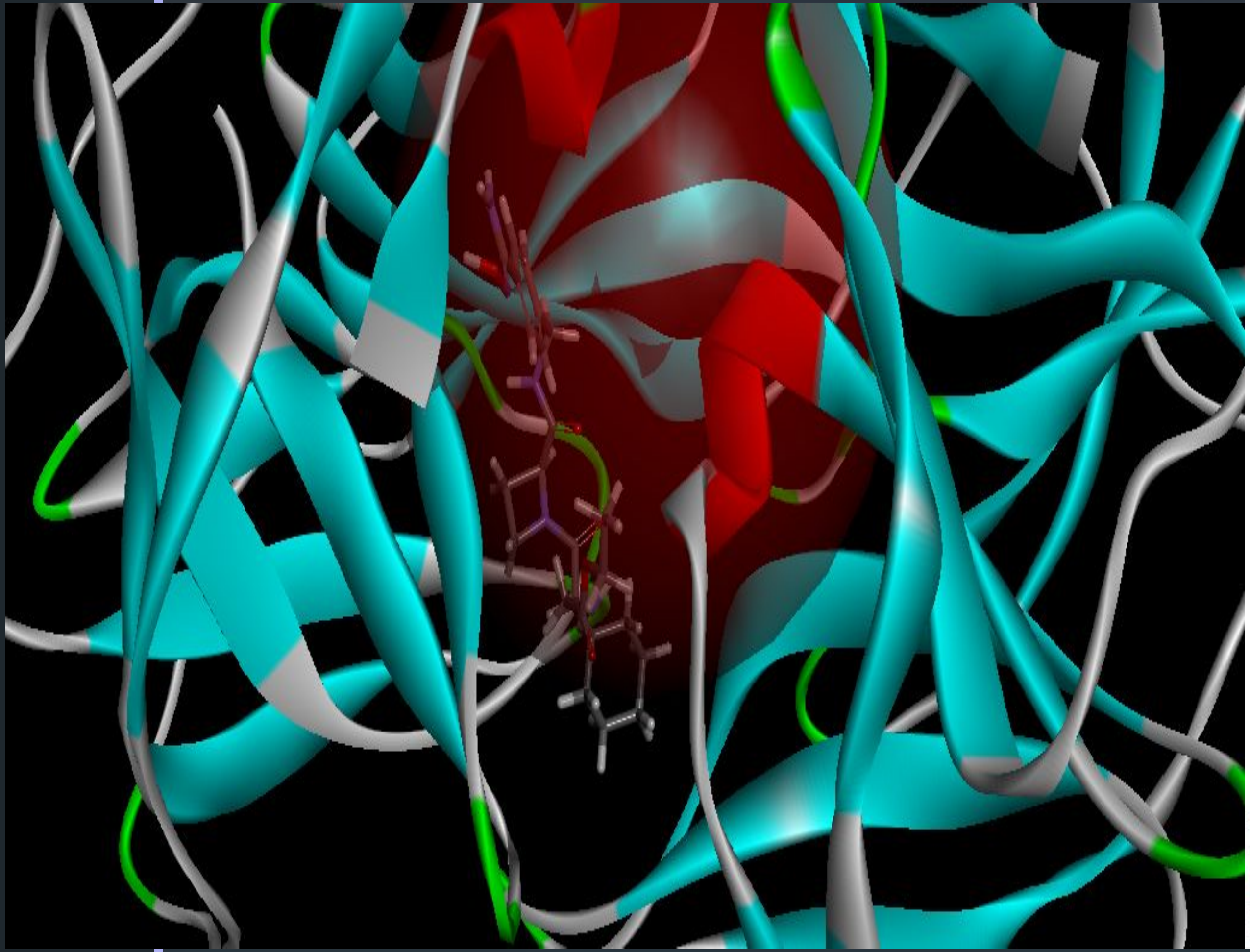
Ximelagatran is not a suitable candidate for treating Zika virus infection due to several reasons:

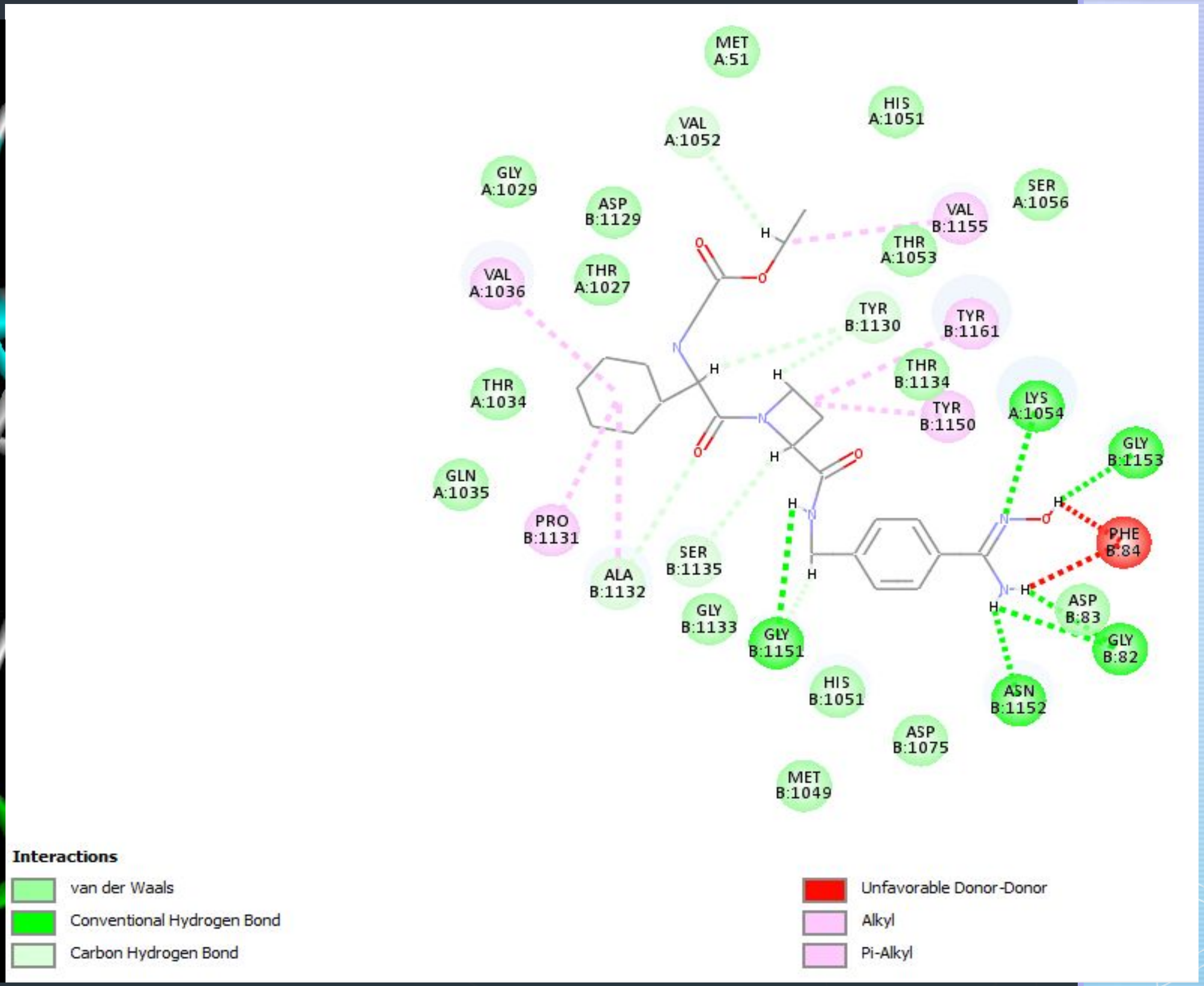
Mismatched mechanisms: Ximelagatran works by inhibiting thrombin, a protein involved in blood clotting. Zika virus, on the other hand, is a single-stranded RNA virus. Thrombin isn't involved in the virus's life cycle, so Ximelagatran wouldn't target the virus directly.

Safety concerns: Research on Ximelagatran was discontinued due to increased bleeding risks associated with its blood clotting inhibition properties.



Interaction of compound and serine protease N3 residues:





References:-

1. Pubchem: https://pubchem.ncbi.nlm.nih.gov/
2. Drugbank: https://go.drugbank.com/
3. POCKDRUG : https://pockdrug.rpbs.univ-paris-diderot.fr/cgi-bin/index.py?page=home
4. CASTP:http://sts.bioe.uic.edu/castp/index.html?1bxw

Thank you!!